The research ecosystem

arXiv, publication, and how knowledge earns trust

Understanding arXiv

arXiv, pronounced 'archive,' is a digital repository for scholarly articles in physics, mathematics, computer science, quantitative biology, quantitative finance, statistics, electrical engineering and systems science, and economics. It serves as a platform for researchers to share their work before or after peer review, making it a crucial part of the academic communication process. However, it's important to note that arXiv itself does not perform peer review or endorse the conclusions of the papers it hosts. A paper on arXiv can also be published in a peer-reviewed journal or conference proceedings, which adds another layer of scrutiny.

arXiv hosts papers without peer review; papers may still undergo external peer review elsewhere. Hosting does not imply peer review status.

The research lifecycle

The research lifecycle includes several stages: preprints, conferences, journals, reviews, standards, and industry papers. Preprints are early versions of papers that have not yet been peer-reviewed. Conferences often feature peer-reviewed proceedings, which are collections of papers presented at the conference. Journals provide a more rigorous peer review process, ensuring the quality and validity of the research. Reviews, such as surveys and systematic reviews, summarize and evaluate existing research, while meta-analyses combine data from multiple studies to draw broader conclusions. Standards and industry papers provide practical guidelines and insights from real-world applications.

Evaluating a paper

When evaluating a paper, it's crucial to look beyond the publication venue and consider the content itself. Peer review is a critical step in ensuring the quality of research, but it's not foolproof. Replication studies, where researchers attempt to reproduce the results of a study, provide additional validation. To assess a paper's credibility, consider the methodology, data quality, and whether the conclusions are supported by the evidence. Also, check if the paper has been cited by other researchers and if there are any critiques or follow-up studies.

Relevance vs. truth

It's important to distinguish between the relevance of a paper to your interests and its truthfulness. A paper might be highly relevant to your work but still contain errors or unverified claims. Conversely, a paper might be less relevant but rigorously peer-reviewed and reliable. Always seek out multiple sources and consider the context in which the research was conducted. Understanding the nuances of the research lifecycle helps in making informed judgments about the credibility and applicability of research findings.

What your site tells us

A limited audit of the retained local implementation - not a verification of every claim

Observed facts and inferences

Your site contains a curated collection of research papers and summaries, but it's essential to distinguish between observed facts and inferences. Observed facts are verifiable pieces of information, such as the title, author, and publication date of a paper. Inferences, on the other hand, are interpretations or conclusions drawn from the data. For example, labeling a paper as 'peer-reviewed' based on its presence on arXiv is an inference, as arXiv does not perform peer review. Understanding this distinction helps in assessing the reliability of the information presented.

A narrative synthesis lacks systematic review's rigorous search and selection criteria. A systematic review must document its methods for reproducibility.

Strengths and limitations

The strengths of your site include its comprehensive collection of research papers and summaries, which provide valuable insights into various topics. However, there are limitations to consider. The site lacks detailed information on the peer review status of papers, and there is no clear methodology for selecting and evaluating sources. Additionally, the site does not provide full-text access to all papers, which can limit the depth of analysis. These limitations highlight the need for a more transparent and rigorous approach to curating and presenting research.

The code cannot show if Claude or humans read, selected, or verified papers; detailed logs and prompts are necessary for such verification.

Meaning of evidence-strength labels

Evidence-strength labels on your site indicate the quality and reliability of the research. These labels are based on factors such as the type of study, the presence of peer review, and the consistency of the findings. However, it's important to understand that these labels are not a guarantee of truth or accuracy. They provide a framework for evaluating the credibility of the research, but they do not replace the need for critical analysis and verification.

Proposed methodology

To improve the credibility and transparency of your site, a proposed methodology could include defining a research question, recording databases and search terms used, and specifying inclusion criteria. This would involve distinguishing between preprints, reviewed papers, policy documents, and industry sources. Reading and summarizing original relevant passages, logging evidence and limitations, and mapping important claims to specific sources and versions would provide a robust foundation. Publishing syntheses with uncertainty and correction dates, and requiring human sign-off for public claims, would enhance the reliability of the information presented. This approach would ensure that the site remains a valuable resource while maintaining high standards of research integrity.

Sources: [1] arXiv - About
[2] arXiv - Identifiers and versions
[3] PRISMA 2020 - Review reporting guidance

Audit basis: local website data and research-backend schema/implementation, inspected 30 October 2026. Drafted locally using Ollama; not published.